JEE Mathematics: Quick Review with Grading and Practice Map - Logarithm

Grade shows how often the item is tested in the DPP questions (104 questions): High = 9 or more questions, Medium = 1 to 8 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the DPP questions that test the item. 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Exponential Form

Question 01

What is exponential form? Explain all its components with their conditions.

Model Answer

$N = a^x$: N = number, a = base, x = power (index). Conditions: $N > 0,\ a > 0,\ a \neq 1,\ x \in \mathbb{R}$.

Ex: $2^x = 8 \Rightarrow x = 3$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

How do you solve an equation by converting to the same base and comparing the powers?

Model Answer

Write both sides with the same base $a > 0,\ a \neq 1$; then $a^{f(x)} = a^{g(x)} \Rightarrow f(x) = g(x)$.

Ex: $16^x = 32 \Rightarrow 2^{4x} = 2^5 \Rightarrow x = \dfrac{5}{4}$.

Grade - Medium

Practice map, main concept: DPP: 4(3)

Also used in: none

Logarithmic Form

Question 03

What is logarithmic form and what are the conditions on its components?

Model Answer

$a^x = N \iff \log_a N = x$, with $N > 0,\ a > 0,\ a \neq 1$ (number positive, base positive and not 1).

Ex: $f(x) = \log_{(x-2)}(5 - x)$ is defined when $x - 2 > 0,\ x - 2 \neq 1,\ 5 - x > 0$, i.e. $x \in (2, 5) - \{3\}$.

Grade - Medium

Practice map, main concept: none

Also used in: DPP: 1(18); 2(1, 6)

Question 04

How do you convert logarithmic form to exponential form?

Model Answer

$\log_a N = x \iff N = a^x$ (base of the log becomes the base of the power).

Ex: $\log_2(x - 1) = 3 \Rightarrow x - 1 = 2^3 \Rightarrow x = 9$.

Grade - Medium

Practice map, main concept: DPP: 1(9)

Also used in: DPP: 1(5, 12-14); 2(4)

Properties of Logarithm

Question 05

What is the value of a log when the number and the base are the same?

Model Answer

$\log_a a = 1$ (for $a > 0,\ a \neq 1$).

Ex: $\log_7 7 = 1$.

Grade - **Medium**

Practice map, main concept: none

Also used in: DPP: 1(15)

Question 06

What is the value of a log when the number and the base are reciprocals of each other?

Model Answer

$\log_a \dfrac{1}{a} = -1$ and $\log_{1/a} a = -1$.

Ex: $\log_5 \dfrac{1}{5} = -1, \quad \log_{1/5} 5 = -1$.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 07

What is the value of a log when the number is 1?

Model Answer

$\log_a 1 = 0$ for every valid base a; also $\log_a(a^p) = p$.

Ex: $\log_{10} 1 = 0, \quad \log_3 81 = \log_3 3^4 = 4$.

Grade - **Medium**

Practice map, main concept: none

Also used in: DPP: 1(15)

Question 08

What is the fundamental log identity and what is its use?

Model Answer

$a^{\log_a N} = N$ $(N > 0)$. Use: $a^x = N \Rightarrow x = \log_a N$; it also removes an exponent that is a log of the same base.

Ex: $2^x = 3 \Rightarrow x = \log_2 3$.

Grade - **Medium**

Practice map, main concept: DPP: 2(15); 3(16)

Also used in: DPP: 1(1, 19); 2(13); 3(9)

Question 09
How do you club two log terms given as a sum or difference?

Model Answer
$\log_a m + \log_a n = \log_a(mn)$, $\log_a m - \log_a n = \log_a \frac{m}{n}$ (same base, $m, n > 0$).

Ex: $\log_{10} a + \log_{10} b - \log_{10} c = 2 \Rightarrow \frac{ab}{c} = 100$.

Grade - High
Practice map, main concept: DPP: 4(28)
Also used in: DPP: 1(6, 11, 20); 2(8-9); 3(1-4, 6-7, 12, 17, 19-20, 22-23); 4(2, 14, 23)

Question 10
What happens to the power of the number of a logarithm?

Model Answer
$\log_a(m^p) = p \log_a m$ for $m > 0$. For even p and any real $m \neq 0$: $\log_a m^2 = 2 \log_a |m|$.
Ex: $\log_2 32^2 = 2 \log_2 32 = 10$.

Grade - High
Practice map, main concept: DPP: 1(10, 15); 4(23)
Also used in: DPP: 1(7, 9, 20); 2(11-15, 24, 26, 28-29); 3(1, 3, 16, 18, 20, 23, 25); 4(14, 28)

Question 11
What happens to the power of the base of a logarithm?

Model Answer
$\log_{a^p} m = \frac{1}{p} \log_a m$. In particular $\log_{1/a} m = -\log_a m$.

Ex: $\log_4 8 = \frac{1}{2} \log_2 8 = \frac{3}{2}$.

Grade - High
Practice map, main concept: DPP: 1(20); 3(24); 4(5, 24)
Also used in: DPP: 1(13); 2(7, 23, 27); 3(2, 4, 7-8, 13, 16); 4(21, 26, 28)

Question 12
How do you club several log terms given in sum/difference form?

Model Answer
First bring every term to one base and put the coefficients as powers (items 10, 11), then club: sums become products, differences become quotients.
Ex: $\log_{10} a - \log_{100} b = 3 \Rightarrow \log_{10} \frac{a}{\sqrt{b}} = 3 \Rightarrow \frac{a}{\sqrt{b}} = 1000$.

Grade - Medium
Practice map, main concept: DPP: 4(2)
Also used in: DPP: 1(10); 2(10-12, 25); 4(5)

Base Changing Theorem

Question 13
Write all four forms of the base changing theorem and explain their applications.

Model Answer

$$\frac{\log_c a}{\log_c b} = \log_b a, \quad \log_c a = (\log_b a)(\log_c b), \quad \log_a b = \frac{1}{\log_b a}, \quad a^{\log_c b} = b^{\log_c a}$$

Use: a chain collapses, e.g. $(\log_2 3)(\log_3 4)(\log_4 5) = \log_2 5$; and $49^{\log_7 5} = 5^{\log_7 49} = 25$.

Grade - High
Practice map, main concept: DPP: 1(1, 3, 6, 8, 11-12, 14, 16); 4(1, 19, 26)
Also used in: DPP: 1(17, 19-20); 2(30); 4(18, 20, 22)

Question 14
What happens to a logarithm when we take its reciprocal?

Model Answer
Number and base are interchanged: $\dfrac{1}{\log_a b} = \log_b a$.

Ex: $(\log_2 3)(\log_3 2) = 1$.

Grade - High
Practice map, main concept: DPP: 1(2, 19)
Also used in: DPP: 1(3, 8, 16); 2(20, 23, 28); 3(5-6, 8-9, 15, 22); 4(13, 18-22, 26, 28)

Question 15
How do you change the base of a log term to another base?

Model Answer
$\log_a b = \dfrac{\log_c b}{\log_c a}$ for any convenient base $c > 0,\ c \neq 1$.

Ex: $\log_8 32 = \dfrac{\log_2 32}{\log_2 8} = \dfrac{5}{3}$.

Grade - Medium
Practice map, main concept: DPP: 1(13, 17); 4(13)
Also used in: DPP: 1(3, 16); 4(15, 20, 29)

Question 16
What is the condition on the number and the base for the value of a log to be positive?

Model Answer
Number and base lie on the same side of 1 (both > 1 or both in $(0, 1)$) $\Rightarrow \log_a b > 0$.

Ex: $\log_7 4 > 0$ and $\log_{2/3} \dfrac{5}{7} > 0$.

Grade - Medium
Practice map, main concept: none
Also used in: DPP: 4(15)

Question 17
What is the condition on the number and the base for the value of a log to be negative?

Model Answer
Number and base lie on different sides of $1 \Rightarrow \log_a b < 0$.

Ex: $\log_{2/3} 5 < 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Antilogarithm

Question 18

How do you convert an antilog term to exponential form and compute its value?

Model Answer

$\log_a N = x \iff \text{antilog}_a(x) = N = a^x$.

Ex: $\text{antilog}_4(3) = 4^3 = 64$.

Grade - Low

Practice map, main concept: none

Also used in: none

Characteristic and Mantissa

Question 19

How do you find the characteristic and mantissa of a number with respect to a given base?

Model Answer

Write $\log_{10} N = c + m$ with $c \in \mathbb{Z}$ (characteristic) and $0 \leq m < 1$ (mantissa). For a negative value, c is the integer just below it.

Ex: $\log_{10} 2^{-50} = -50(0.3010) = -15.05 \Rightarrow c = -16,\ m = 0.95$.

Grade - Low

Practice map, main concept: none

Also used in: none

Number of Digits and Number of Zeroes

Question 20

How do you convert a number into scientific notation?

Model Answer

$N = p \times 10^n$ with $n \in \mathbb{Z}$ and $p \in [1, 10)$.

Ex: $0.00034 = 3.4 \times 10^{-4}$.

Grade - Medium

Practice map, main concept: none

Also used in: DPP: 1(7)

Question 21

How do you find the number of digits of a number using logarithm?

Model Answer

Number of digits $= c + 1$, where c is the characteristic of $\log_{10} N$.

Ex: $\log_{10} 2^{100} = 30.10 \Rightarrow c = 30$, so 2^{100} has 31 digits.

Grade - Medium

Practice map, main concept: DPP: 1(7)

Also used in: none

Question 22
How do you find the number of zeroes after the decimal point using logarithm?

Model Answer
For $0 < N < 1$: number of zeroes after the decimal $= |c+1|$, where c is the characteristic of $\log_{10} N$.
Ex: $\log_{10} 5^{-100} = -69.90 \Rightarrow c = -70$, zeroes $= 69$.

Grade - Low
Practice map, main concept: none
Also used in: none

Log Equation Type 1

Question 23
How do you solve basic log equations by removing the log?

Model Answer
$\log_a m = p \Rightarrow m = a^p$; $\log_a m = \log_a n \Rightarrow m = n$. Club terms first and finally reject every root that makes a number ≤ 0 or a base ≤ 0 or $= 1$.
Ex: $\log_2(x+1) + \log_2(x-1) = 3 \Rightarrow x^2 - 1 = 8 \Rightarrow x = \pm 3$; $x = -3$ rejected, so $x = 3$.

Grade - High
Practice map, main concept: DPP: 1(5, 18); 2(1-3, 6-8, 10-12); 3(2-4, 13-14, 20, 23, 25)
Also used in: DPP: 1(17); 2(5, 9, 15); 3(10-12, 19, 21, 24); 4(24, 29)

Log Equation Type 2

Question 24
How do you use substitution in log equations?

Model Answer
If $\log_a x$ (or a^x) occurs in a repeated or squared form, put $t = \log_a x$ (or $t = a^x$), solve the equation in t, then return to x; remember $a^x > 0$.
Ex: $(\log_3 x)^2 - 3\log_3 x + 2 = 0 \Rightarrow t = 1, 2 \Rightarrow x = 3, 9$.

Grade - High
Practice map, main concept: DPP: 2(4-5, 9, 13, 25-30); 3(1, 5-12, 15, 17-19, 21-22); 4(6, 14, 16, 18, 20, 25)
Also used in: DPP: 2(14, 16-22, 24); 3(13); 4(1, 4-5, 12-13, 19, 29)

Log Equation Type 3

Question 25
How do you solve equations of the form $(\text{variable})^{\text{variable}}$?

Model Answer
Take log of both sides so the powers come down: $f^g = h \Rightarrow g \log f = \log h$; then substitute $t = \log x$ (the variable must keep $f > 0$).
Ex: $x^{\log_{10} x} = \dfrac{100}{x} \Rightarrow t^2 = 2 - t \Rightarrow t = 1, -2 \Rightarrow x = 10, \dfrac{1}{100}$.

Grade - High
Practice map, main concept: DPP: 2(14, 16-24); 4(4, 22, 29)
Also used in: none

Question 26

What is the special case of $(\text{variable})^{\text{variable}}$?

Model Answer

$a^b = a^c \Rightarrow a = 1$ or $b = c$ (for $a > 0$), because $(b - c)\log a = 0$.

Ex: $x^{x^2} = x^{3x-2} \Rightarrow x = 1$ or $x^2 = 3x - 2$, so $x = 1, 2$.

Grade - Low

Practice map, main concept: none

Also used in: none

Logarithmic Inequalities

Question 27

How do you remove the log in inequalities?

Model Answer

$a > 1$: $\log_a M > \log_a N \iff M > N > 0$ and $\log_a M > p \iff M > a^p$. $0 < a < 1$: $\log_a M > \log_a N \iff 0 < M < N$ and $\log_a M > p \iff 0 < M < a^p$.

Ex: $\log_2(x - 1) < 3 \Rightarrow 0 < x - 1 < 8 \Rightarrow x \in (1, 9)$.

Grade - Medium

Practice map, main concept: DPP: 4(17)

Also used in: none

Question 28

How do you solve log inequalities when the base is a variable?

Model Answer

Split into two cases: base > 1 (keep the inequality) and $0 <$ base < 1 (reverse it); in each case also keep the number > 0; take the union of the two case answers.

Ex: $\log_{(x+3)}(x^2 - x) < 1 \Rightarrow x \in (-3, -2) \cup (-1, 0) \cup (1, 3)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Modulus Function

Question 29

What is $|x|$ and what is its value for different values of x?

Model Answer

$|x| = \sqrt{x^2}$ = distance of x from 0: $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.

Ex: $|3| = 3,\ |-4| = 4,\ |0| = 0$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 30
How do you define $|f(x)|$?

Model Answer
Find where $f(x) = 0$ and split the line there: $|f(x)| = f(x)$ where $f \geq 0$ and $|f(x)| = -f(x)$ where $f < 0$.

Ex: $|x - 3| = x - 3$ for $x \geq 3$ and $3 - x$ for $x < 3$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 31
What happens when we square $|x|$?

Model Answer
$|x|^2 = x^2$ and $\sqrt{x^2} = |x|$.

Ex: $x^2 - 5|x| + 4 = 0 \Rightarrow |x| = 1, 4 \Rightarrow x = \pm 1, \pm 4$.

Grade - Medium
Practice map, main concept: DPP: 4(11)
Also used in: DPP: 1(18)

Question 32
Write down the other properties of the modulus function.

Model Answer

$$|x| \geq 0, \quad |x| = |-x|, \quad |xy| = |x||y|, \quad |x+y| = |x|+|y| \iff xy \geq 0, \quad |x-y| = |x|+|y| \iff xy \leq 0$$

Ex: $|x + 2| = -2$ has no solution since $|x| \geq 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Modulus Equation Type 1

Question 33
How do you solve equations of the form $|f(x)| = \text{constant}$?

Model Answer
For a constant $k \geq 0$: $|f(x)| = k \Rightarrow f(x) = k$ or $f(x) = -k$.

Ex: $\left|\dfrac{x+1}{x-2}\right| = 4 \Rightarrow x = 3$ or $x = \dfrac{7}{5}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 34
What do you do for $|f(x)|$ = constant if the constant is negative?

Model Answer
No solution, because $|f(x)| \geq 0$ (solution set $\varnothing$).
Ex: $|x+5| = -1$ has no solution.

Grade - Medium
Practice map, main concept: none
Also used in: DPP: 4(11)

Modulus Equation Type 2

Question 35
What should we keep in mind when we square both sides of an equation?

Model Answer
$A = B \Rightarrow A^2 = B^2$ but not conversely, so squaring can bring extra roots: always check each root in the original equation (a side that equals a modulus must be ≥ 0).
Ex: $|x-3| = 2x-1 \Rightarrow x = -2, \frac{4}{3}$; $x = -2$ makes $2x-1 < 0$, rejected; $x = \frac{4}{3}$.

Grade - Medium
Practice map, main concept: none
Also used in: DPP: 2(25, 27)

Question 36
How and when do you use squaring in equations involving a modulus?

Model Answer
$|f| = |g| \Rightarrow f^2 = g^2$ (no extra root). For $|f| = g$ square only after noting $g \geq 0$, then verify.
Ex: $|x-3| = |2x-1| \Rightarrow 3x^2 + 2x - 8 = 0 \Rightarrow x = -2, \frac{4}{3}$.

Grade - Low
Practice map, main concept: none
Also used in: none

Modulus Equation Type 3

Question 37
How do you solve modulus equations by defining the modulus?

Model Answer
Mark the points where each modulus expression is 0, solve in every interval with the sign of that interval, and keep only roots lying in their own interval.
Ex: $|x-1| + |x-3| = 6$: $x < 1$ gives $x = -1$; $1 \leq x < 3$ gives $2 = 6$ (none); $x \geq 3$ gives $x = 5$.

Grade - Medium
Practice map, main concept: DPP: 4(8, 10, 12, 27)
Also used in: none

Modulus Inequalities Type 1

Question 38

How do you solve inequations of the form $|f(x)| <$ or $>$ constant?

Model Answer

For $k > 0$: $|f| < k \iff -k < f < k$ (intersection); $|f| > k \iff f > k$ or $f < -k$ (union).

Ex: $|x| < 3 \Rightarrow -3 < x < 3$.

Grade - Medium

Practice map, main concept: DPP: 4(9)

Also used in: none

Question 39

What do you do for $|f(x)| <$ or $>$ constant when the constant is negative?

Model Answer

$|f| < k$ with $k < 0$: no solution. $|f| > k$ with $k < 0$: true for every x where f is defined.

Ex: $|x| < -3 \Rightarrow \varnothing$; $|x| > -3 \Rightarrow x \in \mathbb{R}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Modulus Inequalities Type 2

Question 40

When can we square in inequalities and what should we keep in mind?

Model Answer

Square only when both sides are ≥ 0: then $A < B \iff A^2 < B^2$. Then use $A^2 - B^2 = (A - B)(A + B)$ and a sign scheme.

Ex: $|x - 3| < |2x - 1| \Rightarrow (3x - 4)(x + 2) > 0 \Rightarrow x \in (-\infty, -2) \cup \left(\frac{4}{3}, \infty\right)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 41

How and when do you use squaring in inequations involving a modulus?

Model Answer

Use it for $|f| < |g|$ or $|f| > |g|$ (both sides non-negative): $|f| < |g| \iff f^2 < g^2$.

Ex: $\left|\frac{x-1}{x-3}\right| < 2 \Rightarrow (x-1)^2 < 4(x-3)^2$, $x \neq 3 \Rightarrow x < \frac{7}{3}$ or $x > 5$.

Grade - Low

Practice map, main concept: none

Also used in: none

Modulus Inequalities Type 3

Question 42

How do you solve modulus inequations by defining the modulus?

Model Answer

Same as the equation method: split at the zeros of the modulus expressions, solve in each interval, intersect with the interval, then take the union of the pieces.

Ex: $|x-1|+|x-3|<6 \Rightarrow x \in (-1,5)$.

Grade - Medium

Practice map, main concept: DPP: 4(7)

Also used in: none

Exponential Inequalities

Question 43

How do you compare the powers on both sides of an inequality?

Model Answer

Same base a: $a>1$: $a^x > a^y \iff x > y$; $0<a<1$: $a^x > a^y \iff x < y$.

Ex: $2^x < 8 = 2^3 \Rightarrow x < 3$.

Grade - Low

Practice map, main concept: none

Also used in: none

Irrational Inequalities

Question 44

How do you solve the different types of irrational inequalities?

Model Answer

$\sqrt{f} < g \iff f \geq 0,\ g > 0,\ f < g^2$. $\sqrt{f} > g \iff [f \geq 0,\ g < 0]$ or $[g \geq 0,\ f > g^2]$. $\sqrt{f} \geq 0$ always (where defined).

Ex: $\sqrt{x+2} > x \Rightarrow x \in [-2,2)$; $\sqrt{x+2} < x \Rightarrow x \in (2,\infty)$.

Grade - Low

Practice map, main concept: none

Also used in: none

Miscellaneous

Question 45

How do you compare two logarithmic numbers (decide $>$, $<$ or $=$)?

Model Answer

Bring both to the same base (items 11, 15), or compare each with 1 using the sign rule (items 16, 17); for $t + \frac{c}{t}$ type sums use AM-GM.

Ex: $\log_5 7 > 1$ and $\log_7 5 < 1$, so $\log_5 7 > \log_7 5$.

Grade - Medium

Practice map, main concept: DPP: 4(15, 21)

Also used in: none

Question 46

How do you simplify a nested radical $\sqrt{a \pm 2\sqrt{b}}$?

Model Answer

Find p, q with $p + q = a,\ pq = b$; then $a \pm 2\sqrt{b} = (\sqrt{p} \pm \sqrt{q})^2$ and $\sqrt{a \pm 2\sqrt{b}} = \sqrt{p} \pm \sqrt{q}$ (larger root first for the minus sign).

Ex: $12 \pm 6\sqrt{3} = (3 \pm \sqrt{3})^2$, so $\sqrt{12 + 6\sqrt{3}} + \sqrt{12 - 6\sqrt{3}} = 6$.

Grade - **Medium**

Practice map, main concept: DPP: 1(4)

Also used in: none